

MELBOURNE LAVERTON, AustraliaWMO: **948650**Lat: **37.8564S**Lon: **144.7567E**Elev: **20**StdP: **101.08**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.8	2.9	-0.3	3.7	11.2	0.8	4.0	9.6	12.5	12.8	11.4	12.8	2.8	310	0.569

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.3	35.4	19.1	32.5	18.5	29.6	18.1	21.2	28.4	20.2	27.6	19.4	26.5	6.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.2	14.0	23.1	18.1	13.0	22.4	17.2	12.3	21.6	61.4	28.4	58.2	27.6	55.4	26.6	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
11.4	10.2	9.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-0.7	40.8	1.0	2.5	-1.4	42.6	-2.0	44.1	-2.6	45.5	-3.3	47.3		
			-1.0	23.0	1.0	0.9	-1.7	23.6	-2.3	24.1	-2.9	24.6	-3.6	25.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	20.0	20.1	18.1	15.0	12.4	10.2	9.8	10.3	11.9	13.9	16.2	17.9	(6)
	DBStd	4.79	3.91	3.59	3.47	3.02	2.60	2.16	2.05	2.31	2.86	3.39	3.63	3.84	(7)
	HDD10.0	93	0	0	0	1	5	23	30	24	9	2	0	0	(8)
	HDD18.3	1618	22	18	46	110	183	243	266	249	194	144	87	54	(9)
	CDD10.0	1782	309	283	252	149	81	29	22	33	67	124	186	246	(10)
	CDD18.3	263	73	67	40	8	1	0	0	0	2	8	22	42	(11)
	CDH23.3	2931	774	640	408	93	3	0	0	1	16	151	302	543	(12)
	CDH26.7	1344	405	296	171	22	0	0	0	0	2	49	122	277	(13)
(14) Wind	WSAvg	4.7	5.0	4.8	4.6	4.1	4.2	4.2	4.7	4.8	5.0	4.9	4.9	5.0	(14)
(15) Precipitation	PrecAvg	547	45	41	30	48	46	37	39	45	48	53	53	48	(15)
	PrecMax	809	141	272	121	200	123	97	81	89	128	165	133	115	(16)
	PrecMin	295	3	0	5	8	5	10	10	11	14	5	4	2	(17)
	PrecStd	128	33	51	20	40	27	19	19	18	24	31	31	29	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	37.1	35.6	29.3	23.1	18.0	17.8	20.8	25.8	31.7	33.8	38.0	(19)
		MCWB	20.0	19.4	19.2	16.5	14.1	12.6	11.9	13.3	14.8	16.2	18.1	19.3	(20)
	2%	DB	35.3	33.7	31.2	25.9	20.5	16.5	15.9	18.1	22.0	27.4	30.2	33.8	(21)
		MCWB	19.3	19.4	18.3	15.6	14.0	12.2	11.2	12.1	13.9	15.7	17.5	18.2	(22)
	5%	DB	31.3	30.5	27.9	23.4	18.8	15.3	14.7	16.4	19.8	24.2	26.8	29.5	(23)
		MCWB	19.0	19.0	17.7	15.3	13.3	11.6	10.5	11.2	13.1	14.8	16.9	17.7	(24)
	10%	DB	27.1	27.1	24.8	20.9	17.2	14.2	13.6	14.9	17.9	21.2	23.6	25.3	(25)
		MCWB	18.7	18.7	17.3	14.8	12.8	11.0	10.0	10.5	12.2	13.8	16.0	17.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	22.2	21.0	18.1	16.1	14.1	12.9	14.0	16.3	18.2	20.6	21.0	(27)
		MCDB	29.9	30.0	29.5	25.0	20.0	16.5	16.0	19.2	22.2	25.7	28.3	28.3	(28)
	2%	WB	21.3	21.0	19.7	17.1	14.8	12.9	11.8	12.7	14.7	16.7	19.0	19.7	(29)
		MCDB	28.8	27.6	26.6	22.0	18.7	15.4	14.8	17.2	20.2	23.9	26.1	27.9	(30)
	5%	WB	20.2	20.1	18.8	16.2	14.0	12.1	11.0	11.7	13.6	15.4	17.8	18.7	(31)
		MCDB	28.1	26.8	24.9	21.1	17.7	14.6	13.8	15.4	18.9	22.2	24.5	26.4	(32)
	10%	WB	19.2	19.3	17.9	15.3	13.2	11.4	10.4	10.9	12.6	14.3	16.6	17.7	(33)
		MCDB	26.5	25.5	23.6	20.0	16.6	13.8	13.1	14.3	17.1	20.2	22.6	24.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.3	10.1	9.8	8.4	7.5	7.6	8.6	10.0	11.0	10.5	11.0	(35)
		MCDBR	19.4	17.8	16.8	14.5	10.4	8.7	8.6	10.7	13.1	17.1	17.8	19.8	(36)
	5% WB	MCWBR	6.2	6.0	6.3	6.4	5.6	5.5	5.1	5.9	6.7	7.5	7.2	6.8	(37)
		MCWBR	14.6	13.2	12.6	11.4	9.1	7.7	7.7	9.9	11.8	14.5	14.5	14.8	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.352	0.357	0.331	0.332	0.315	0.302	0.303	0.316	0.333	0.338	0.349	0.344	(40)
		taud	2.505	2.499	2.573	2.537	2.569	2.593	2.581	2.530	2.483	2.486	2.482	2.514	(41)
	Ebn at Noon	Ebn at Noon	981	952	938	872	829	813	835	876	920	960	977	993	(42)
		Edh at Noon	112	108	93	84	71	64	69	83	99	108	114	112	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	7.15	6.24	4.96	3.45	2.29	1.78	2.00	2.82	4.03	5.37	6.29	7.06	(44)
	RadStd	RadStd	0.45	0.32	0.28	0.20	0.13	0.14	0.11	0.18	0.28	0.42	0.38	0.40	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
(46)	Station Only	(d) +0.60	(e) +0.36	(f) +0.54	(g) N/A	(h) +0.40	(i) +0.47	(j) -29	(k) -151	(l) +199	(m) +72	(46)
(47)	Regional (38 neighbors)	+0.51	N/A	N/A	+0.87	N/A	N/A	-17	-129	+175	+59	(47)

Nomenclature: See separate page